



The broad-band impedance budget in the accumulator ring of ALS-U project

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North American Particle Accelerator Conference
1-6 September, 2019
Lansing, Michigan

Acknowledgment

AR Impedance Collaboration:

- Physics group: Marco Venturin, Stefano De Santis, Derun Li, Tianhuan Luo (LBNL)
- Engineering group: Michael Mardenfeld, Tom Miller (LBNL)
- Karl Bane (consultant) and Robert Warnock from SLAC
- Rui Li from Jlab

Outline

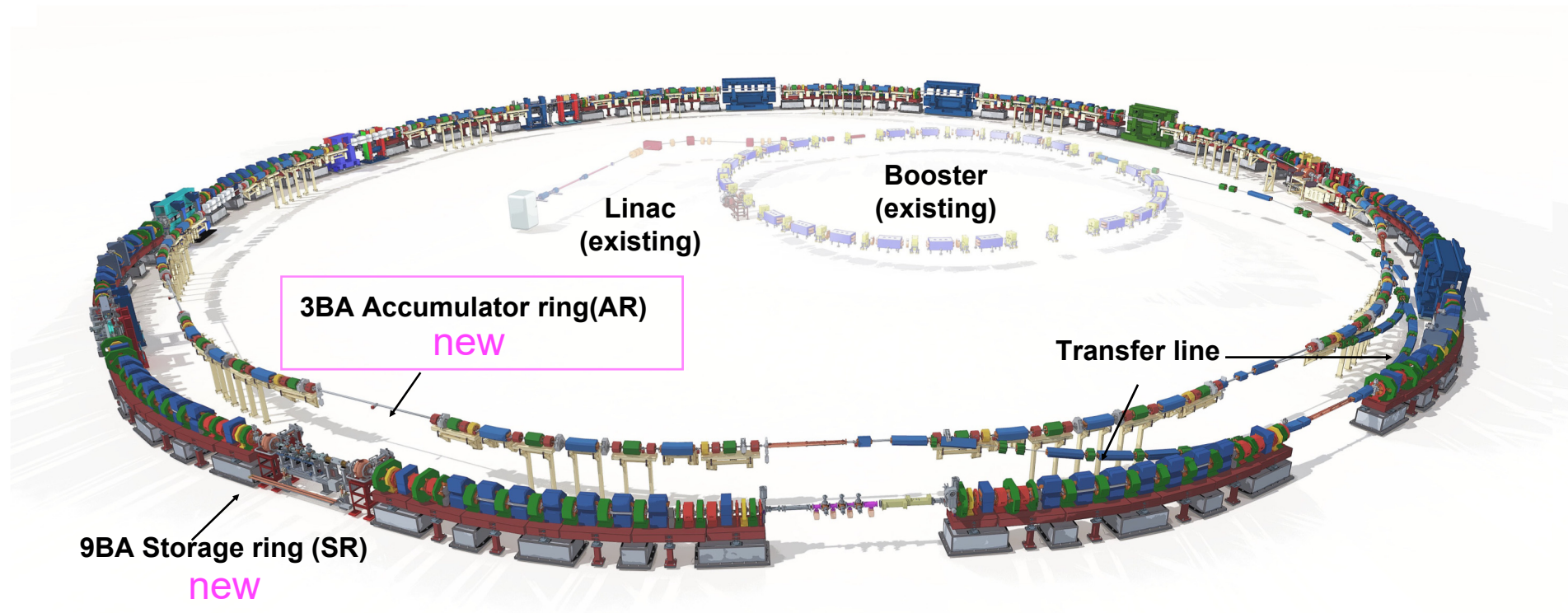
1. Motivation
2. Broad band Impedance modeling
 - Resistive wall
 - Geometric
3. Total budget: longitudinal & transverse
4. Beam instability study due to broad band impedance
 - Longitudinal threshold
 - Transverse threshold
5. Discussion and Summary

Motivation



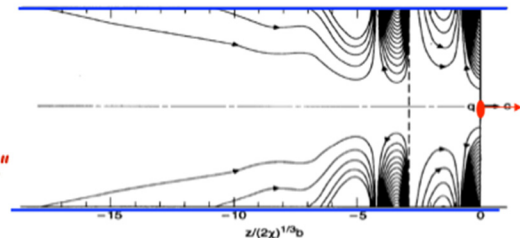
Introduction of the ALS-U project

- Upgrade of the Advanced Light Source (ALS-U) to a multi-bend-achromat lattice as a 4th generation LS
- Provide a soft x-ray source that is up to 100–1,000 times brighter than today's ALS

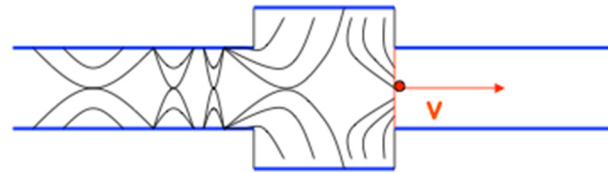


Short-range wakefield/BB impedance in the ring

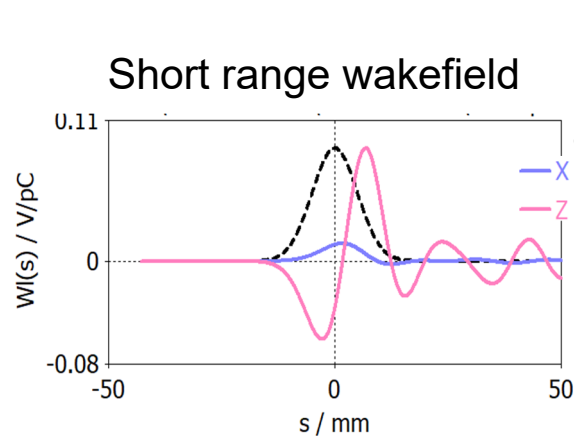
- Source: resistive wall // geometric elements
- Effect : deterioration of beam quality // overheat the vacuum chamber ☹️



Smooth pipe with finite resistance



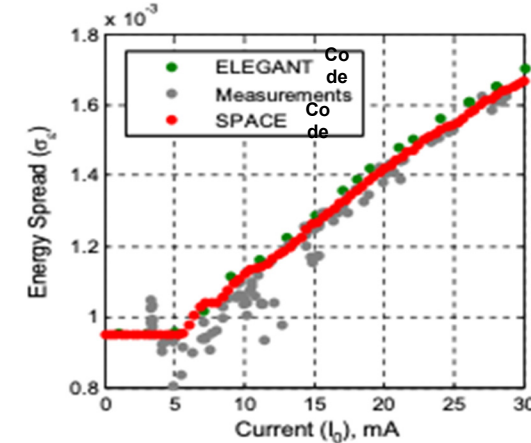
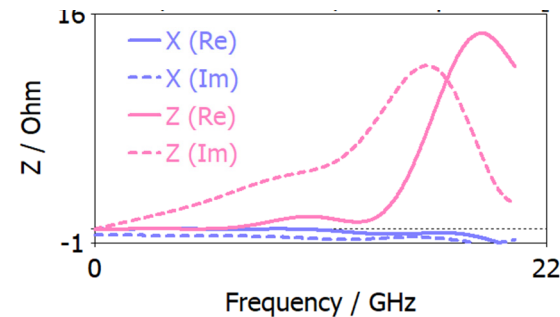
intersection of a geometry change



Short range wakefield

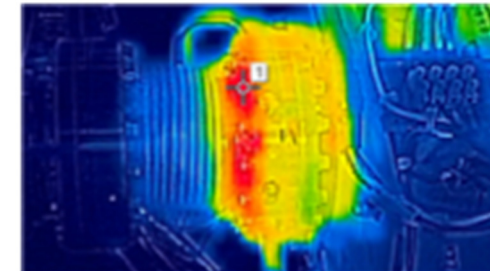
FFT

Broad-band Impedance



Energy spread increase when charge goes high due to BB impedance @ APS

A. Blednykh, et., al.,
NAPAC2016, TUPOB51



Local overheating image of flange joint @ NSLS-II

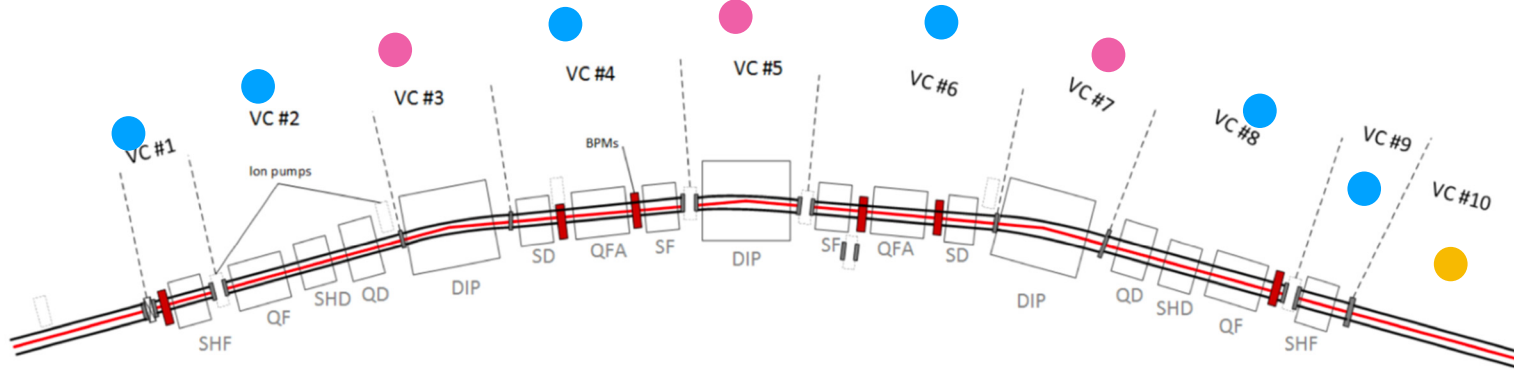
A. Blednykh, et., al.,
IPAC2019, TUPGW082

Impedance modeling



Identify the impedance source_1: resistive wall

- Resistive wall impedance is calculated with analytical formulas



1 normal sector in the AR, 12 sectors in total

	Type	radius (mm)	Profile	Length (m)	Material
●	Dipole	7 (y) 20(x)	ellipse	3	Aluminum
●	Arc	14	round	7.8	Stainless steel
●	Straight	23.6	round	4.2	Stainless steel

$$Z_{||}(\omega) = \frac{Z_0 \delta \omega}{4\pi b c} (\text{sign}(\omega) - i) \times F_{YOKOYA}$$

$$Z_{\perp}(\omega) = \frac{Z_0 \delta}{2\pi b^3} (\text{sign}(\omega) - i) \times F_{YOKOYA}$$

Chamber radius: b

Skin depth: $\delta = \sqrt{2/\mu_0 \sigma_c |\omega|}$

F_{YOKOYA} : geometric factor, $F_y=0.83$ and $F_x = 0.43$ for AR elliptical dipole chamber with $b = 7$ mm

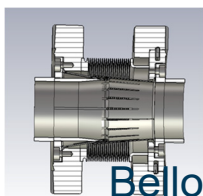
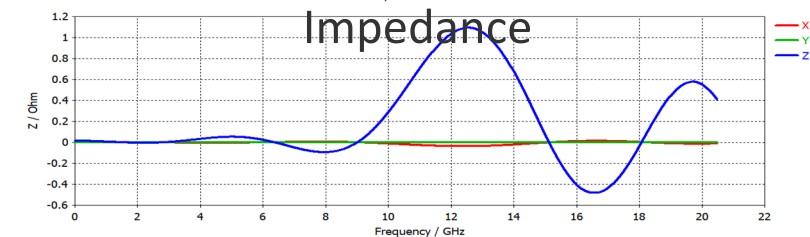
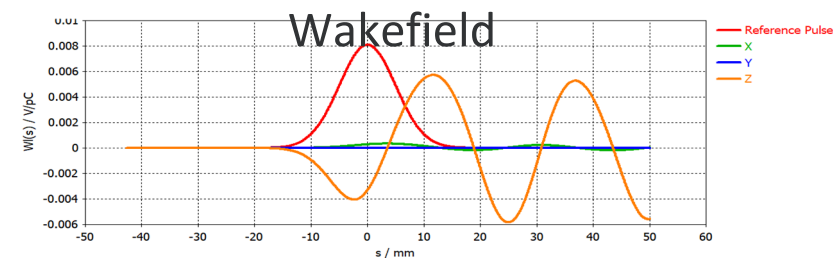
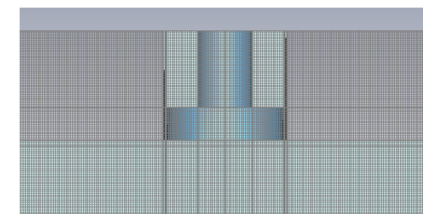
1. A.Piwinski, Wakefield and ohmic losses in round vacuum chambers, 1992
2. K. Bane, The short range resistive wall wakefield, 1996
3. K. Yokoya, in Proceedings of International Conference on Particle Accelerators (IEEE, 1993), pp. 3441–3443

Identify the impedance source_2: geometric elements

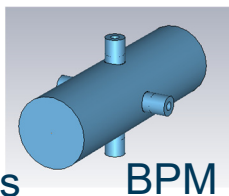
- Majority of geometric elements has been studied
- Geometry impedance is solved by CST Particle Studio

Per sector (12 sectors in AR)		In the ring	
Component	Quantity	Component	Quantity
Flange	20	RF cavity	2
Pump screen	4	Cavity transition	2
Transition_DA	3	LFB kicker	1
Transition_SA	1	LFB transition	1
BPM	6	Stripline kicker	1
Inline pump	4		
Bellow	7		

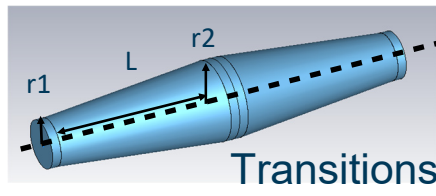
Example: AR
button BPM in
CST with meshing



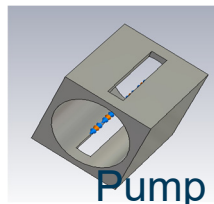
Bellows



BPM



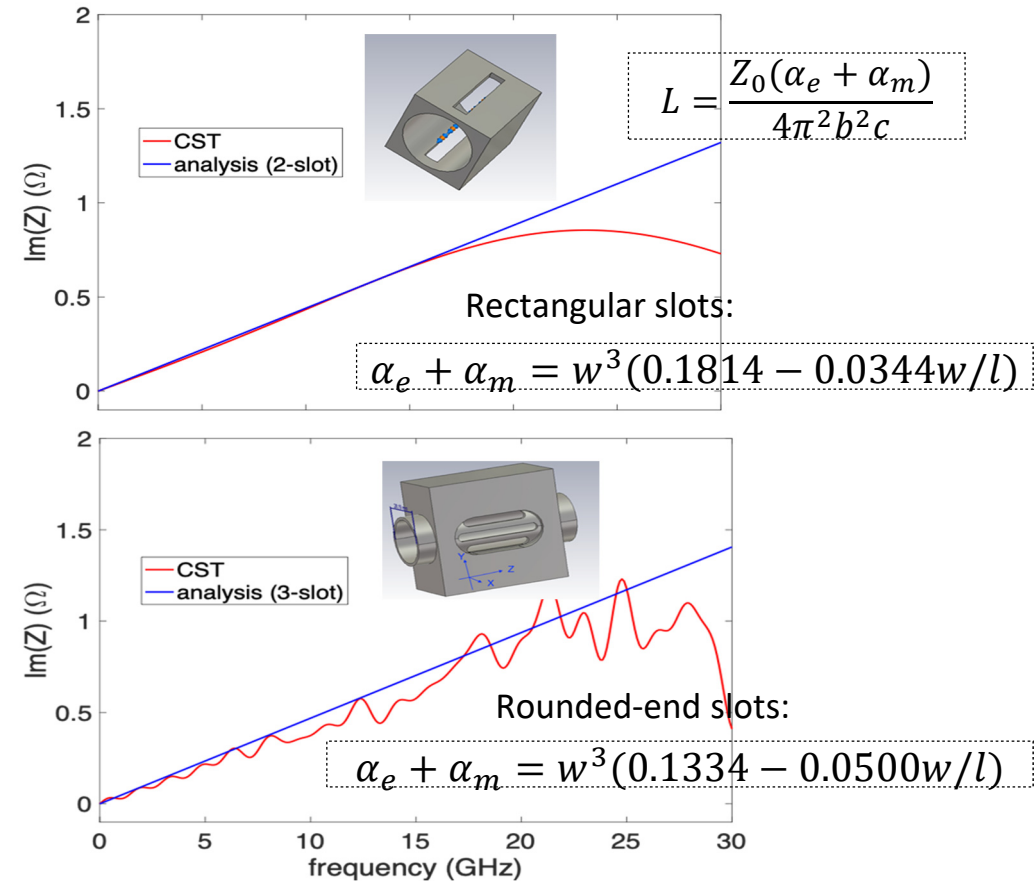
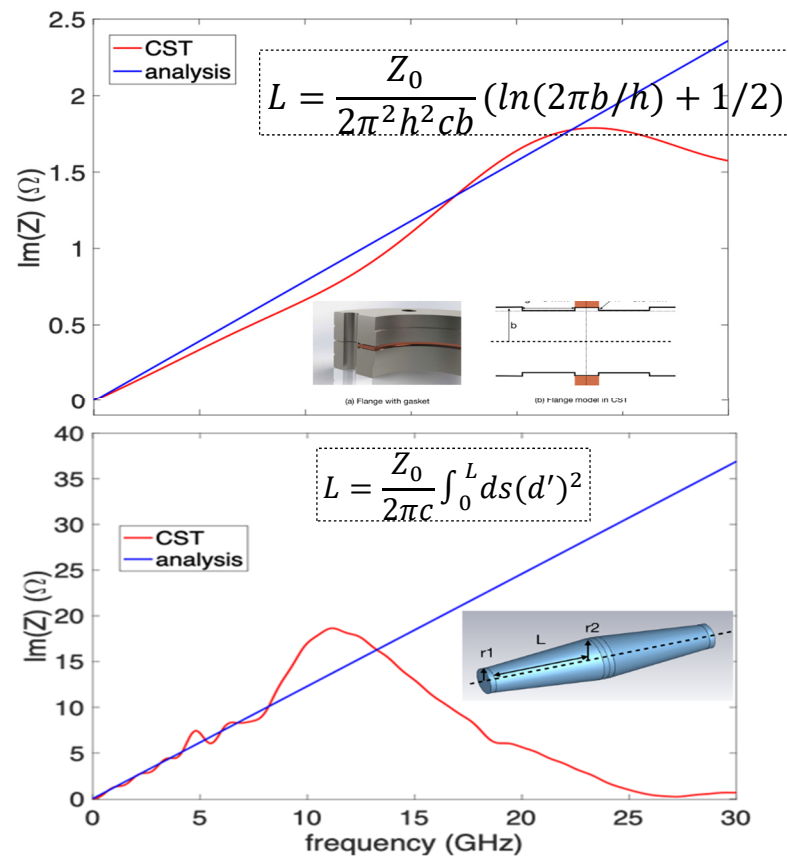
Transitions



Pump screen

Cross-check with analytical formula as much as applicable

- Good agreement on inductive impedance ($Z_{||} = -j\omega L$) at low frequency*



* All formulas above can be found in Accelerator Handbook

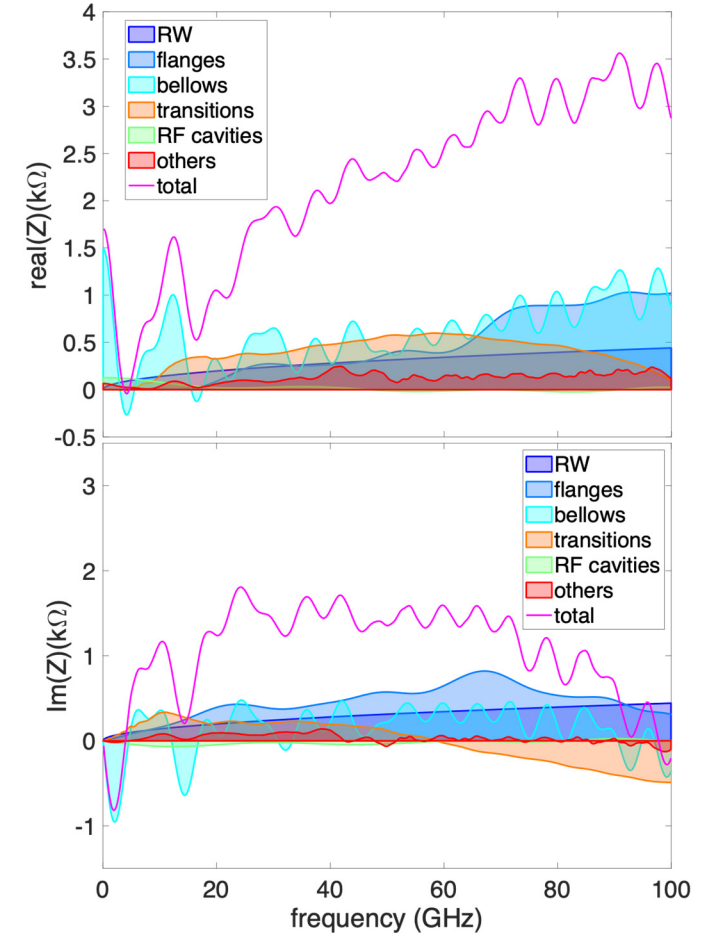
Total impedance budget



Longitudinal budget

- Table: loss factor ($\sigma_z = 5\text{mm}$) and normalized impedance (Z/n , with $n = \omega/\omega_0$)

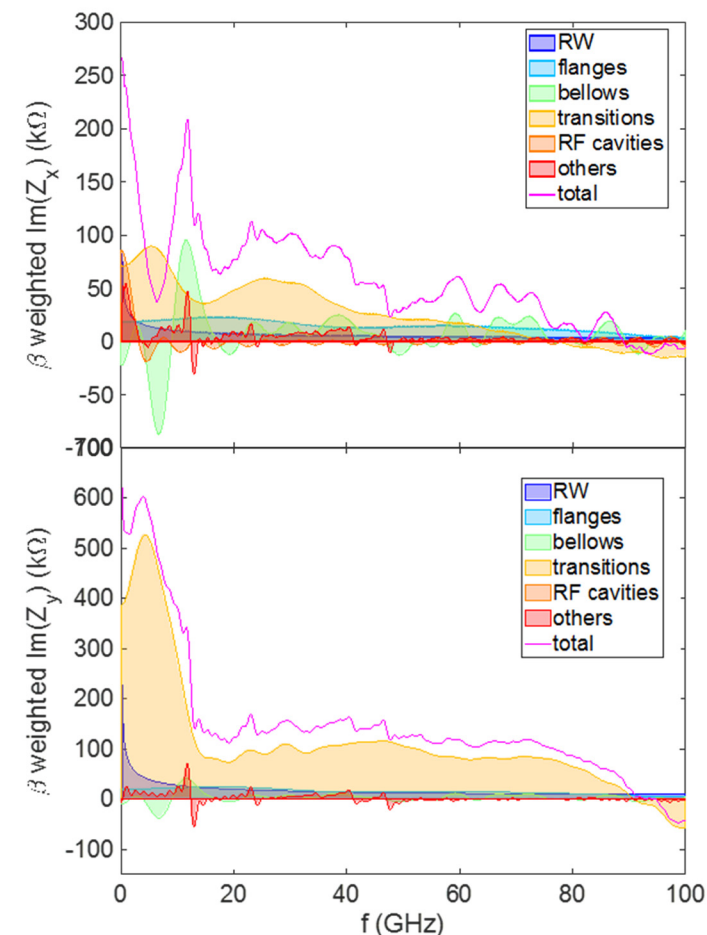
Component	Quantity	Single component			Total		
		Loss factor V/pC	Re(Z/n) m Ω	Im(Z/n) m Ω	Loss factor V/pC	Re(Z/n) m Ω	Im(Z/n) m Ω
Bellow	84	0.097	0.984	0.250	8.148	82.622	21.000
Cavity	2	0.980	9.930	-7.097	1.960	19.859	-14.193
Resistive wall	1	1.600	16.232	24.012	1.600	16.232	24.012
Transition_SA	12	0.041	0.413	2.109	0.492	4.959	25.311
LFB kicker	1	0.490	4.969	-3.383	0.490	4.969	-3.383
Transition_AD	36	0.004	0.018	0.871	0.144	0.660	31.369
BPM	72	0.0014	0.014	0.040	0.101	1.022	2.872
Cavity transition	2	0.088	0.898	1.973	0.176	1.796	3.946
Flange	240	0.00034	0.013	0.118	0.082	3.129	28.414
LFB transition	1	0.075	0.765	1.620	0.075	0.765	1.620
Arc pump screen	24	0.00058	0.006	0.129	0.014	0.142	3.104
Straight Pump screen	24	0.00057	0.006	0.098	0.014	0.139	2.340
Stripline kicker	1	0.0104	0.087	~ 0.000	0.010	0.087	~ 0.000
Inline pump	48	0.00012	0.001	0.032	0.006	0.059	1.545
Ring Total					13.3	136.4	128.0



Transverse budget

- Table: β -weighted effective impedance and tune shift ($Q=1.15nC$, $\sigma_z = 5mm$)

Component	Quantity	Single component (k Ω)		Total			
		$(\beta * Z_{eff})_x$	$(\beta * Z_{eff})_y$	$(\beta * Z_{eff})_x$ k Ω	$(\beta * Z_{eff})_y$ k Ω	Tune shift x *10 ⁻⁴	Tune shift y *10 ⁻⁴
Transitions_AD	36	0.24	9.69	8.74	348.70	-0.068	-2.699
RW_Arc section	1	15.34	28.42	15.34	28.42	-0.119	-0.220
RW_Dipole chamber	1	1.49	27.74	1.49	27.74	-0.012	-0.215
Transitions_SD	12	4.88	1.90	58.53	22.83	-0.453	-0.177
Flange	240	0.09	0.09	20.66	22.40	-0.160	-0.173
Pump screen	48	0.03	0.36	1.33	17.17	-0.010	-0.133
BPM	72	0.10	0.15	6.88	10.80	-0.053	-0.084
Inline pump	48	0.28	0.12	13.31	5.62	-0.103	-0.043
Bellow*	84	0.12	0.05	10.37	4.49	-0.080	-0.035
Cavity	2	6.92	2.31	13.83	4.61	-0.107	-0.036
Cavity transition	2	2.31	0.77	4.61	1.54	-0.036	-0.012
LFB kicker	1	4.52	1.51	4.52	1.51	-0.035	-0.012
RW_Straight section	1	3.99	1.33	3.99	1.33	-0.031	-0.010
LFB transition	1	2.04	0.68	2.04	0.68	-0.016	-0.005
Stripline kicker	1	0.51	0.17	0.51	0.17	-0.004	-0.001
Ring total				166.18	498.00	-1.29	-3.85



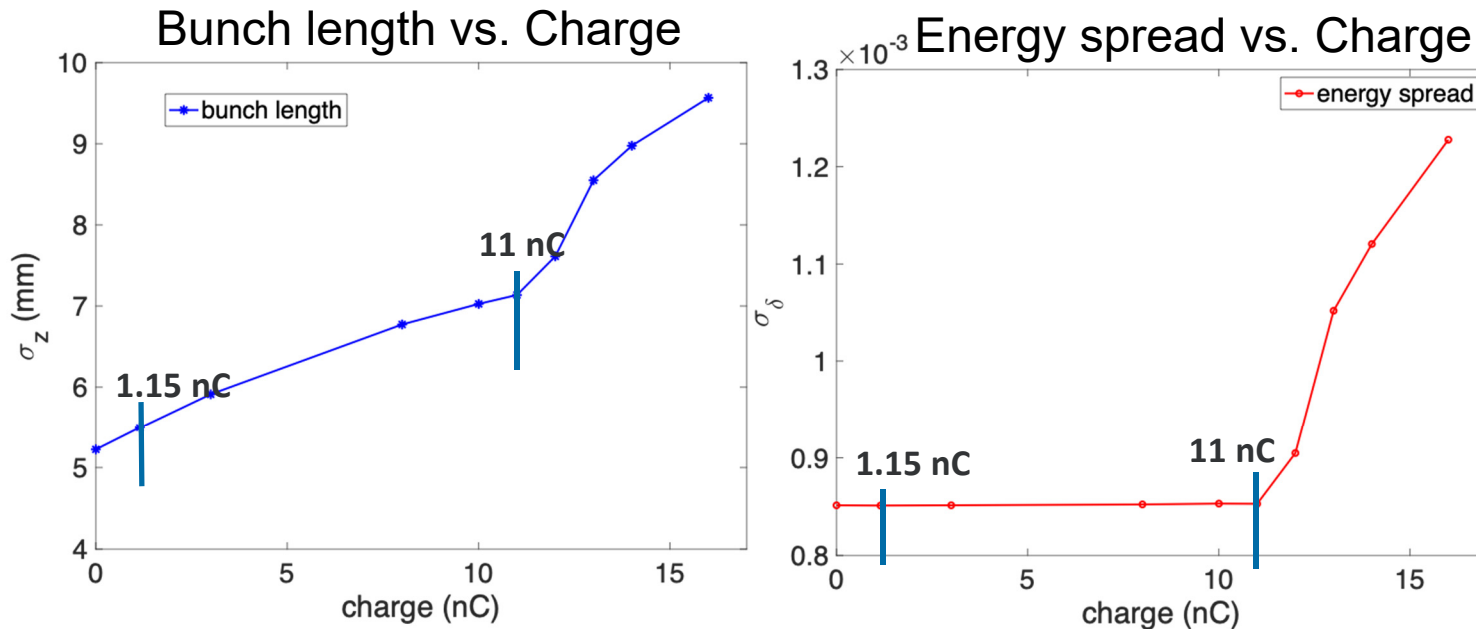
Instability study



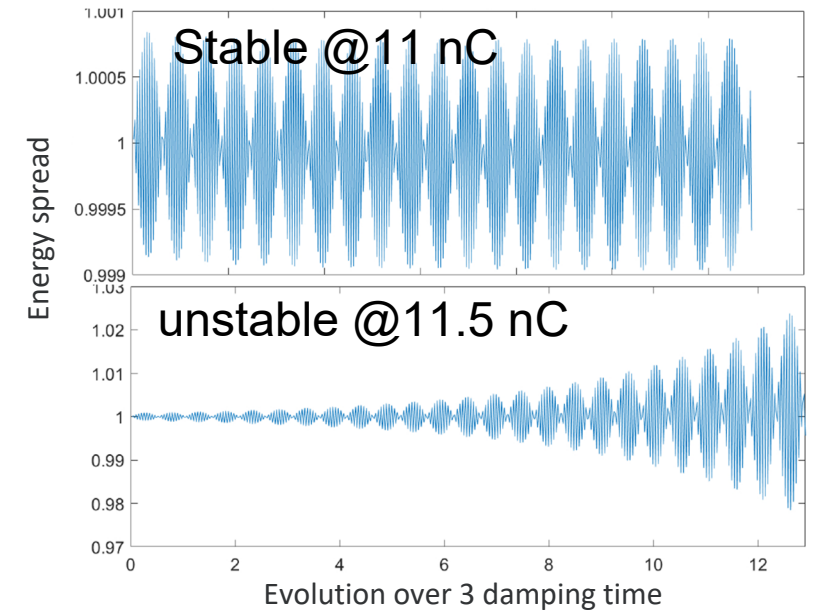
Longitudinal: microwave instability

- ELEGANT simulation: onset of microwave instability at charge of ~ 11 nC
- Results verified by VFP solver

From ELEGANT



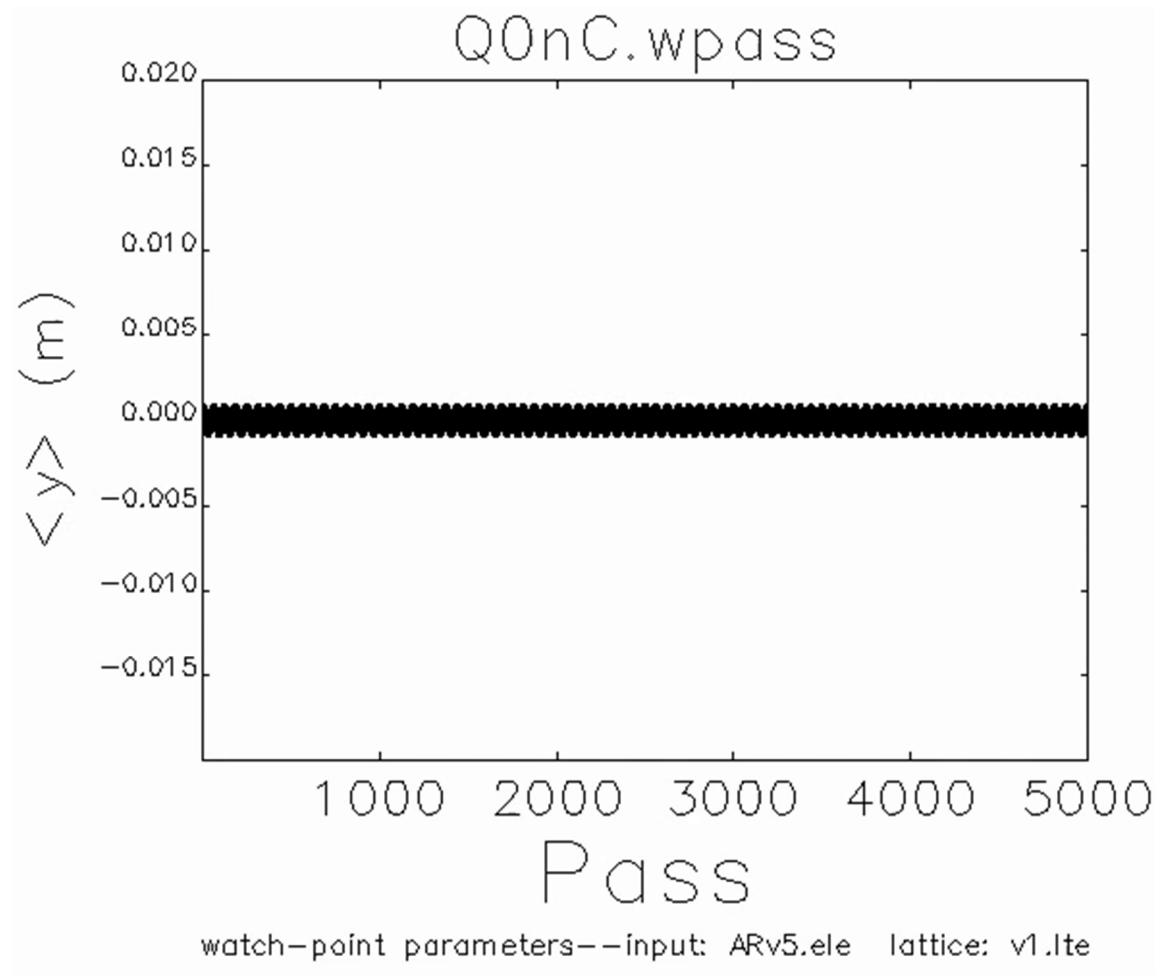
From VFP solver



Courtesy of R. Warnock

Transverse mode coupling instability

- ELEGANT simulation: onset of transverse mode coupling instability at charge of ~12 nC



Summary



Summary & Plan

- Broad-band impedance of ALS-U AR have been extensively surveyed. Majority of the impedance sources have been analyzed.
- Based on the current impedance model, beam dynamics study shows the AR bunch charge (1.15 nC/bunch) is well below the threshold of the longitudinal instability threshold (~ 11 nC/bunch) and transverse instability threshold (~ 12 nC/bunch).
- The total impedance model will be updated as more vacuum components are included, or their design is modified.



Thanks!